



EHRT-RWB: A Novel Ensemble Hybrid Recurrent Transformer for Multimodal Heart Disease Risk Prediction

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Abstract The disease that contains the highest mortality and morbidity across the world is cardiac disease. Annually millions of people are affected and deaths take place due to cardiac diseases worldwide. There are various diagnostic measures for the prediction of heart diseases. However, these techniques consist of some errors, delays, and high-cost consumption. These limitations affect the patients in many ways such as inadequate medication during the right time, affecting their mental health, affecting their physical health, and sometimes death. Hence, there is a need for an effective automatic multimodal disease risk prediction mechanism related to heart diseases. Therefore, this paper proposes the Ensemble hybrid recurrent transformer-based random wolf bird (EHRT-RWB) algorithm. The details of the patient acquired as input by the heart disease dataset are preprocessed initially and classification is performed with the EHRT-RWB methodology. The RNN method extracts reliable features and helps to study the temporal sequence data. Time series classifications are maintained with the help of the transformer method and the ensemble method integrates advanced cutting-edge technologies for performance augmentation. Finally, the RWB algorithm selects and tunes the hyperparameters of the developed classification method. Experiments conducted using certain performance assessment measures and the comparison with existing strategies

show the method achieves superior cardiac disease detection performance with the highest accuracy score of 98.7%.

Keywords Cardiac diseases · Hybrid recurrent neural network · Transformer model · Weighted average ensemble · Wolf-Bird optimizer

Introduction

Recently, people have been affected by various diseases because of the gradual change in the progression of their living habits and environmental factors [1]. Brain diseases, diabetes, visual impairment, heart diseases, and cancer are considered as the major diseases by which the people are most commonly affected. It is stated that nearly 423 million people are affected by diabetes worldwide [2]. Coronary heart diseases and strokes are comprised of cardiovascular disease. It is reported that nearly 19 million pillions lost their lives in 2019. In Europe, most of the people lost their lives only by these cardiovascular diseases [3]. The guidelines are provided by the European Society of Cardiology (ESC) to develop risk prediction models and also to improve the prevention methods [4]. By employing several longitudinal studies and analyzing several autonomous regressions, the prediction model is developed [5].

The powerful tools are provided by the machine learning (ML) algorithms to the researchers. The ML is employed in several medical-related sectors which include the treatment, diagnosis process, interpretation of clinical images, and predicted output [6, 7]. The Cleveland heart disease dataset is utilized by various researchers. Suitable data are required by the predictive model of ML for the testing and training phases [8, 9]. The accuracy of the ML classifiers is improved by using the standard dataset for training and

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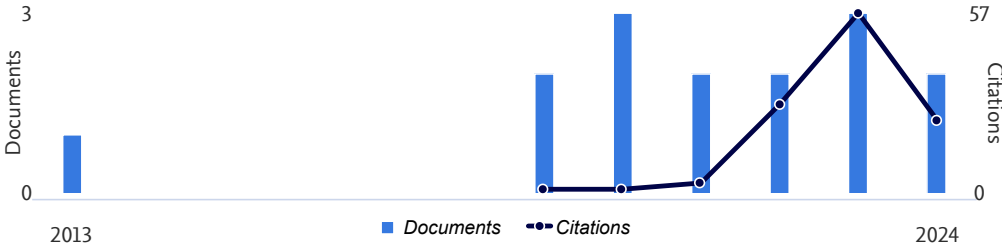
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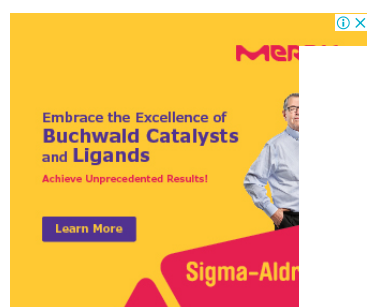
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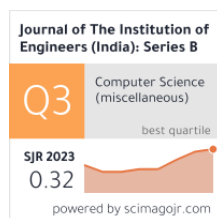
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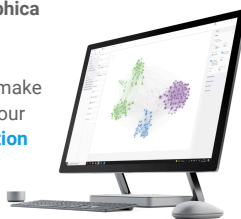
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